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Generative AI Applications For Occupational Safety Engineering: Successful Implementations And Lessons Learned

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Abstract

Occupational safety performance highly relies on integrating lessons from near-misses and past accidents. Artificial Intelligence, and in particular Generative AI, provides significant opportunities to reanalyze massive corpuses of data related to occupational safety, and contributes to the transformation in the safety/HSE engineering domain. This paper examines various approaches implemented since early 2023 using Generative AI.

The internal TotalEnergies digital solution, ‘SafeTE’, has demonstrated to be an effective platform for proving concepts, benchmarking capabilities, and creating new user experiences by leveraging historical safety data.

The solution successfully covers several use cases targeted at different persona and operational processes:

- Learn by experience: Based on the accidents and near-misses database: GenAI summarizes and finds related incidents from accidents and near-misses. This is the backbone of SafeTE. The objective is to access and associate faster cases with operations, to learn from their causes, assess and study the correctives and remediations actions.
- Based on the anomalies, observation cards and stop cards records: topic modelling with GenAI continuously models and tracks trends in field reports. This allows weak signal detection and pair comparison between Operational sites.
- By connecting between planned activities and past accidents/near-misses cases database, GenAI is leveraged to interpret connects the context of planned activities with relevant past incidents occurred in a similar context, to typically support toolbox talks and job preparation.

After over a year of use, the application shows that generative AI is well-suited for occupational safety data enhancement and elaboration of safety performance-oriented use cases. Not only the Large Language Models perform well in this domain (where technical jargon is limited and where situational comprehension is generally accessible to these types of technologies), but user experience is enhanced with respect to previous processes, where reliance on memory was critical for accident recall, as well as a limitation.

The key findings are:

- LLMs / Agents, semantic vectorization and topic modelling do work very well for occupational safety, where technical jargon is not critical to situational understanding.
- Establishing benchmark procedures enables us to track the quality / performance of LLMs over time, as new models become available.
- Process safety remains more challenging as we write this paper. The user research shows it is a very different use case, but and jargon, technicity, lower heterogeneity of situations makes it a still-to-explore domain.

This paper explores the use cases, technical implementation, and lessons learned from applying Generative AI for the integration of textual and document data in occupational safety engineering. The good users and stakeholders adoption confirms the relevance of this innovative technique in defeating information overload and increasing situational relevance.

Introduction

Oil and gas industry has significantly improved its performance in safety indicators over the years. Since its early steps in the beginning of the XXth century, safety engineering has much progressed through a mix of technology and safety management systems (better job preparation, risk assessments, execution procedures). Part of the journey was based on the conviction of the importance of "learning from past incidents" as well as "learning from what weak signals and small anomalies can tell".



Figure 1—Extract of internal pitch for the solution: key pillars of safety engineering

Digital technologies seemed appropriate to offer opportunities to boost those two dimensions.

To deliver pinpointed insights prior a job execution via a digital solution, and to enhance safety engineers' ability to detect weak signals from past company incidents' reports, the authors tried the application of now-conventional Natural Language Processing in 2021 and rapidly abandoned these techniques. Semantic similarity was possible, yet not decisive alone.

The large-scale introduction of GPTs in 2023 reinvigorate the work, led to an increased ambition in the use cases, and accelerated the road to successful implementations, described in this paper.

Available data

The application of safety management systems produces a significant quantity of textual material.

- Incident and near-miss reports (referred to in this paper as data system DS1)
- Anomaly and observations 'cards' (DS2)
- Work permits (WP)
- Lessons learned from incidents ("Return of experience") and Cause Tree Analysis

In our implementation, we added to the internal Incident and near-miss reports the ones from the IOGP (International Oil and Gas Producer Association), that add a high-quality dataset of high severity (whether actual or High Potential near-miss) incidents. Overall, data availability and data quality were proven sufficient and well adapted for the pursued purposes.

STEP1: Safety engineering use cases: User research and product discovery

The starting point of the project was to defeat information overload and enable the identification of weak signals in the data. The main risk identified at project startup was the risk to generate additional information, without providing the right level of insight prescriptively and without adequate user experience / tool simplicity to make an impact.

Pre-existing situation: Learning from past accidents and accident recall

In TTE, high quality and structured data for the last 25 years is available through a robust database system, provided by a sector-leader vendor, that globally has a good reputation. However, in the light of technological opportunities, it appeared that:

- The solution almost exclusively focuses on record-keeping and numerical analytics.
- It is equipped with a classic and deceptive search tool. On the safety engineering users group (HSE engineers, site managers), users were able to find mostly cases they were recalling. This requires an effort and leads to subjectivity. For the same group, finding inspiring cases associated to other sites' history was relatively random. People would frequently prefer to contact colleagues to hear about 'their' relevant cases. Regular manual updates of compilations of 'important to remember' cases were done. In that perspective, the internal safety engineering network is playing a major role in spreading safety alerts and remembering worth-to-learn-from incidents. In short, without a powerful search tool, memory and experience are the main success factors in recalling accidents.
- Access to the solution is global, yet for the reasons exposed previously, the content is of easier access for trained safety engineers than for occasional users or operational managers.

Expressed challenge: Linking pre-work toolbox-talks to relevant accident recalls

Toolbox talks—discussions held on work location before initiating a work activity—are a systematic practice within TTE and many other companies. Multiple stakeholders noted that during work permit reviews conducted the day before a job, it is commonly desired that superintendents are asked to identify a relevant "case-to-be-recalled".

This practice heavily relies on the experience and memory of the participants. To improve its effectiveness, it is essential to associate each upcoming work situation with one or more historical incidents that are contextually more relevant than those already known to the team. However, existing search tools within accident databases often fall short in identifying such contextually relevant cases. This is primarily because incident reports tend to focus on causes and consequences, rather than providing detailed descriptions of the job context in which the incident occurred.

The second need identified, is to identify the relevant "lesson learned" reports ('REX' for return of experience in TTE jargon) during the work preparation itself, in order to ensure to the activity owner that he has covered, anticipated, and is prepared for all the hazards he faced during the work execution.

Pre-existing situation: Observation cards

Employees in all roles - and particularly operational ones - are incentivized to report every anomaly they may observe, or every positive observation associated with the workplace. Positive observations include shoutouts and acknowledgement of good practices being implemented, or relevant interventions. Any report is made via what is called an "observation card".

The success of this type of system is such that it is not rare to have companies collecting hundreds of thousands of records a year. TTE typical affiliates with a few operated sites generate 15 to 25 thousand observation cards a year. It means dozens of cards a day on each site that must be reviewed and treated by site safety engineers and offshore operation crew.



Figure 2—Typical workflow for observation cards

Extracting learnings from this volume of data is associated with 2 specificities:

- The high heterogeneity of situations results in many (non-exclusive) categories or classes of cards, along with many miscellaneous cases. Establishing a consistent and objective classification system requires significant effort and is often described as "difficult to maintain consistently" over time.
- While summarization and synthesis of cases can be performed and reported on site, there is a tendency to anchor on the most cognitively striking cases. This can lead to overlooking weak signals, such as gradual drifts in less impactful categories.

These cards do contribute to resolving situations — Line management and safety supervisors do take action based on them. However, the historical absence of a systematic classification has limited the usability of the data generated through these cards.

- It was difficult to extract and share insights about the content of the cards ("what's in there?") beyond the initial perimeter.
- It was not feasible to interpret a high number of cards as a sign of a concerning safety score on site, since 1/ positive cards are not always properly flagged and 2/ multiple reports from different observers represent a statistically significant portion of the dataset.
- The key concern remains that weak signals are not detected in time.

Remark: By definition, observations refer to situations that did not result in an incident, near-miss, or highly unacceptable outcome. If an anomaly is identified as serious at the time of reporting, it is immediately reclassified and added to the "learning-from-past-incident" database as a near-miss or HIPO (High Potential Anomaly).

Additional Insights and user research findings

- Similar cases: Users frequently seek cases similar to a situation they have in mind. This helps expand the impact of the initial case, reinforce risk assessments, and broaden insights through analog yet distinct scenarios.
- Safety engineers often build their own compilations of cases when preparing for major operations or campaigns.

- There is currently no tool that presents cases in a light, easy-to-digest format with minimal effort—ideally requiring only a few clicks. Such a tool could offer passive exposure to relevant cases, helping maintain high safety awareness. While management and HSE engineers receive detailed case presentations, weekly safety digests, and lessons learned reports, operational crews are typically exposed only to content curated by safety engineers or superintendents.
- Hard to read reports: Incident reports are often difficult to read, especially for those unfamiliar with the context. They may require significant focus and even multiple readings. Summarization and rewriting using large language models (LLMs) have shown promise in improving accessibility and increasing report consumption.
- Cause extraction: Although reporters are encouraged to include causal elements in the database, this is not consistently done—particularly for low-severity cases. This leads to inconsistent classifications, limiting the utility of the data for safety engineering purposes.
- Languages/translations: Occasionally in some sites, English has not always been the primary reporting language, translations are welcome.
- Regarding observations: Safety engineers are interested in evaluating the effectiveness of awareness campaigns (e.g., on PPE use, eye protection, or adherence to golden rules). They aim to determine whether such campaigns lead to a measurable reduction in related observations.
- Emulation: There is a belief that sites compete for the best safety performance. Highlighting objective differences between sites can foster positive discussions and knowledge exchange. It's allowed also site management to detect earlier weak signal with this comparison with Pairs.

Other tools were being developed within the company at the same time, such as a risk assessment assistant—which analyzes work permits and suggests additional risks and mitigation measures—and an advanced search tool for the internal safety referential. These use cases are not covered in this paper, which focuses primarily on two areas: leveraging data to learn from past incidents and analyzing observation card data.

STEP2: Execution – Leveraging the Generative AI wave, with a focus on prescriptivity and user experience

The execution phase was built upon the previously described user research and the capabilities offered by Generative AI and semantic technologies. The resulting tool, internally named SafeTE (pronounced safe-Tee-Hee), consists of three main modules.

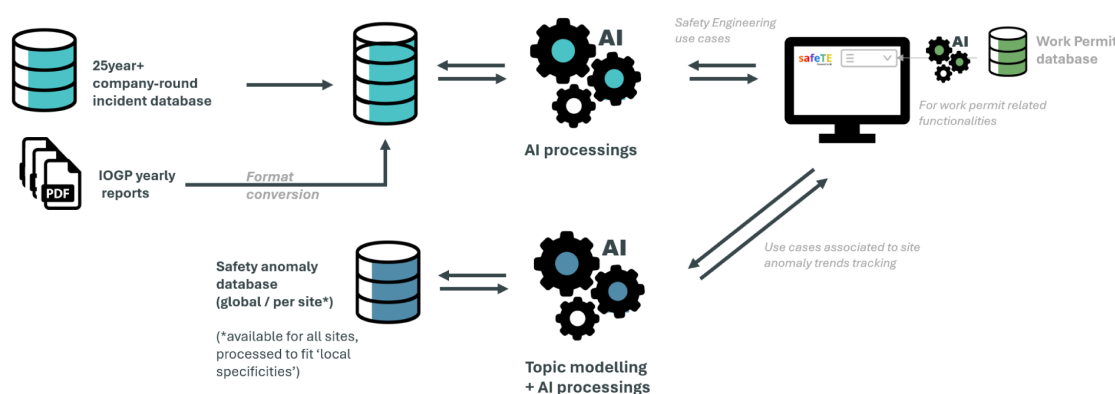


Figure 3—Data flow architecture for the solution

Use Case 1: Exposure, search, and browsing of past incidents and near misses

It aims to clearly present incidents with revised titles, LLM summaries, and suggestions for related cases. This information is presented on a standardized incident card form (Figure 4)

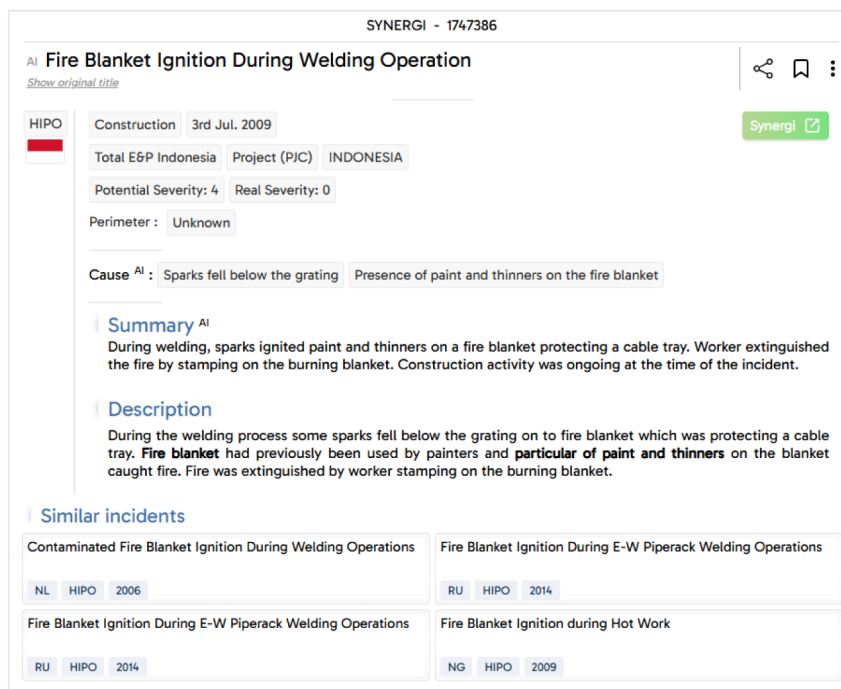


Figure 4—Incident Card in the SafeTE search tool: data exposed is enhanced by AI (cause, enhanced title, summary, similar accident...)

Figure 5 presents the basic flow of how incidents are extracted and processed on a daily basis to prepare the "enhanced incident dataset". The key techniques involved at that stage are:

- LLM-based summarization of incident reports
- LLM-based cause extraction
- Semantic vectorization of documents & Semantic similarity between documents to provide relevant "similar incident lists"

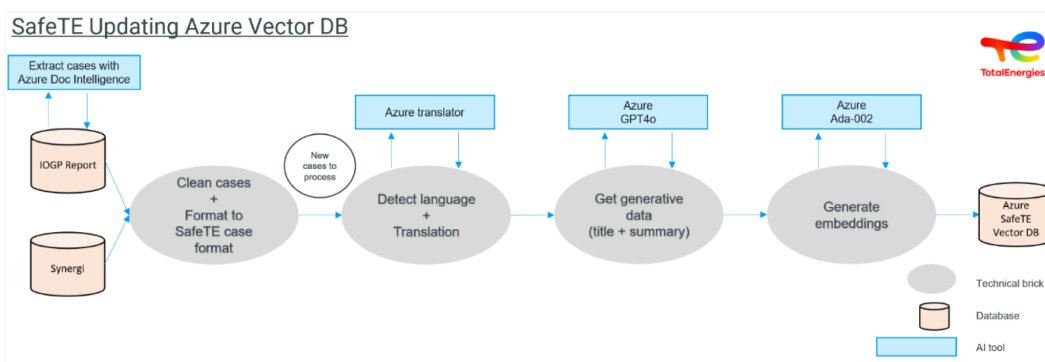


Figure 5—Process to update the Vector Database (Azure AI Search Index)

This module enables users to explore historical safety data through a semantic-based search interface.

After observing the natural behavior of users, we realized that they often mixed in the query elements of dates (periods), locations, with description of causes, consequences or nature of incidents. We also noted some users were referring to 'Golden rules' labels (Golden rules are an equivalent of the IOGP Live Saving rules in TotalEnergies and serve as classification label for most incidents.)

To enhance the search from complex free texts, an AI layer is designed to extract what is better if algorithmically treated as a filter (dates, places) and what is true ‘natural language objective of the search.

Figure 6 shows how the filters such as date and country are automatically extracted from the free text query. In that case the query is “welder burnt in Nigeria between 2014 and 2021”. Other filters like Golden Rule and site would be treated similarly. Based on the query, SafeTE returns incident cards that correspond to the most relevant cases. Results are prioritized per severity and Fatalities are highly ranked.

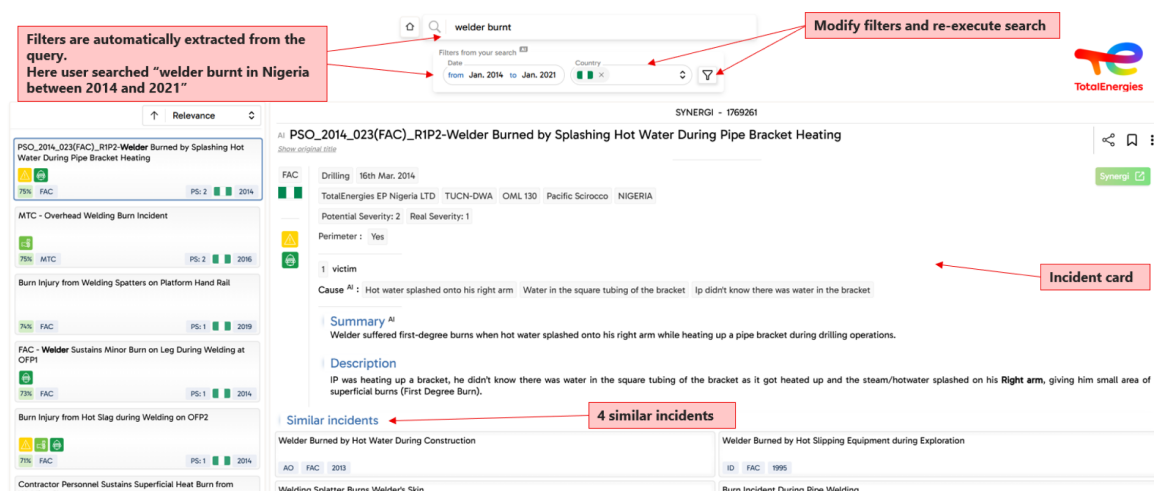


Figure 6—Result of a search for "welder burnt in Nigeria between 2014 and 2021" in the SafeTE search tool. Filters are extracted from the query to ensure the search focuses on the semantic content.

Figure 7 illustrates how a user's query is handled by SafeTE, including the automatic extraction of filters from the free text. The key techniques involved in this search component are:

- The original search query analysis strategy, leveraging filters and transformation of queries for semantic search (referred to as *Odata filters*)
- Semantic vectorization of documents and search

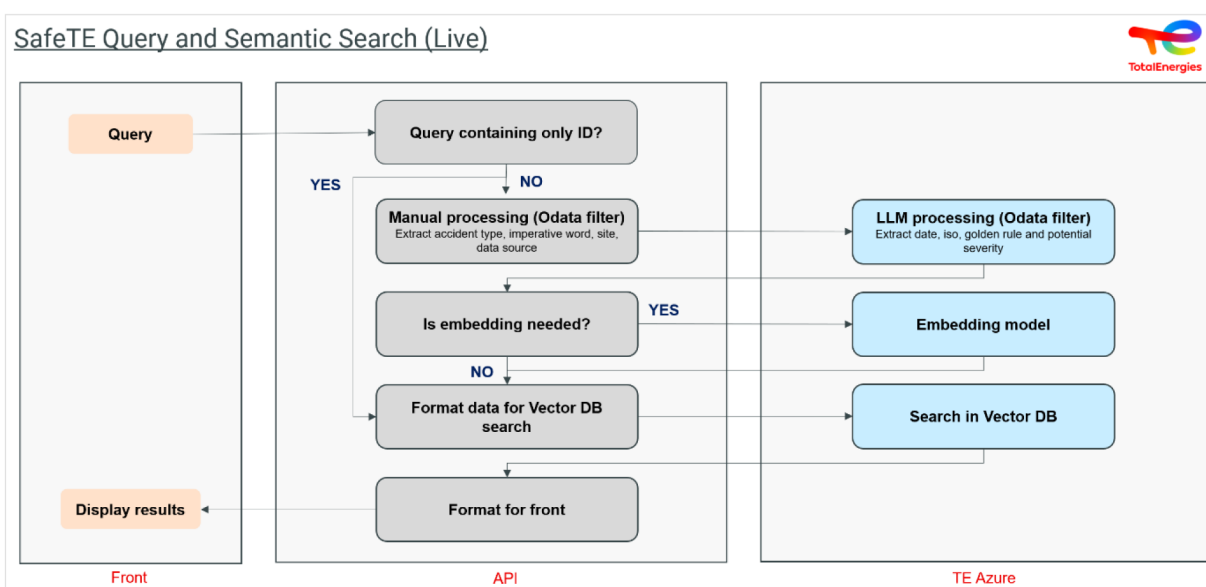


Figure 7—Process flow to treat a live query and generation of Odata filters.

These developed search capabilities and the quality of the results provided have been appreciated by the early adopters over the preexisting search tool embedded in the commercial database software. This module has largely contributed to the expansion of the solution to the operational site teams.

Illustration of the boosting in term of quality of search

One of the first examples we took to compare search between conventional database tool and safeTE was about looking accidents that involved a lifting trolley. Figure 8 shows the results from the embedded search in the original database system, when looking for "lifting trolley". Relevance is very low. Figure 9 shows safeTE results. Note that a relevance score is provided, and the semantic search tool is designed to maximize it.

The screenshot displays a search results page for the query 'lifting trolley'. The interface includes a sidebar with filters, a main results area, and a top navigation bar.

Filters (2):

- Case type: Event - Incident
- Where and What - Case date: 0

Search Results (141 hits):

Case no.	Event - Incident	Status	Description
#677044	EVENT - INCIDENT	CLOSED WITHIN DUE DATE	FAC - Minor Head Injury
#677040	EVENT - INCIDENT	CLOSED AFTER DUE DATE	One employee fell on the floor while moving 20kg CO2 wheel fire extinguisher
#673039	EVENT - INCIDENT	CLOSED WITHIN DUE DATE	FAC - Foot sprain when stepped down from the transport trolley
#671682	EVENT - INCIDENT	NOT DUE	LTI: IP impacted his hand to vertical beam while doing manual handling
#671127	EVENT - INCIDENT	CLOSED WITHIN DUE DATE	ML - Compressor bundle tips and slides off trolley
#670206	EVENT - INCIDENT	CLOSED AFTER DUE DATE	MTC: IP Strained back while transporting scaffold material on trolley
#667135	EVENT - INCIDENT	CLOSED WITHIN DUE DATE	LTI - Deck crew person experienced back pain after several days of manual handling from Haven to TEH (food, equipment, material ...)

Figure 8—Search results for the query 'lifting trolley' in the original database (commercial industry legacy solution)

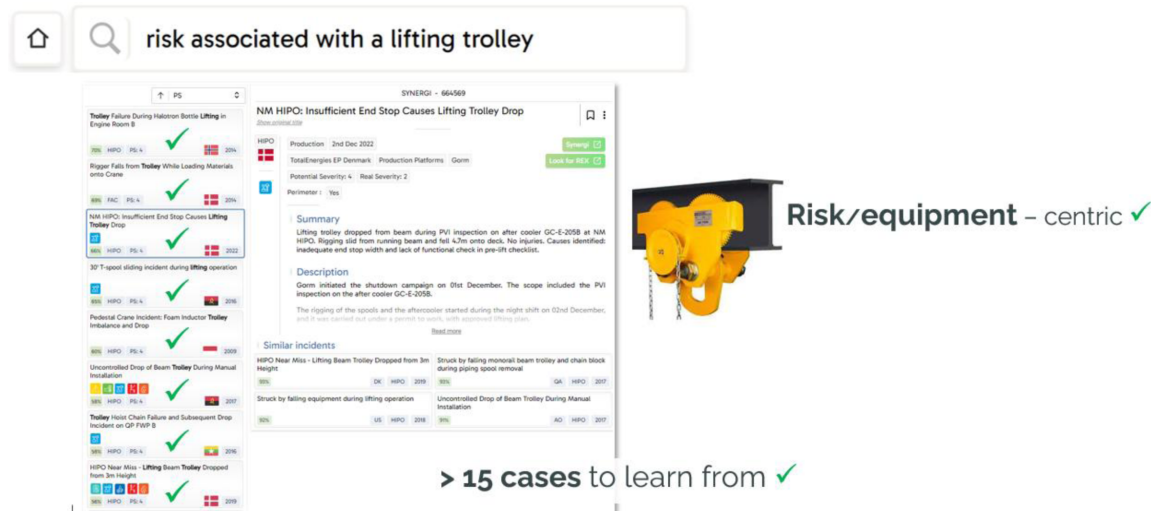


Figure 9—Search results in safeTE semantic search, on the same underlying database as previous figure.

Use Case 2: Recalling relevant incidents based on upcoming work context

This module, called *SafeWork*, supports the preparation of pre-job toolbox talks (TBT) by identifying past incidents that are contextually relevant to the upcoming work, based on the description provided in the work permit. The goal is to connect current risk situations with historical accidents to enhance awareness and prevention. This is particularly valuable for long-duration activities, as a permit can cover up to seven days of activities, making TBTs potentially redundant for workers. SafeWork offers an innovative alternative.

SafeWork is also used to reassure the activity owner or the campaign leader that all risks have been studied and covered during the operation preparation. Reviewing root cause analysis and correctives and mitigation measures implemented after an incident occurred in a similar environment, is giving important insights to the work preparation.

The user can find their permit which has already been pre-processed. This action retrieves dozens of past incidents that should be relevant to the work described in the permit. Figure 10 displays an example permit and the relevant past incidents identified by the tool.

Title: COW25-C4 - DFG Dump Caisson Support Modification. Welding support Description: Repair Scope for welding of new support. Task Steps: > Planning and toolbox talk. > Preparation of worksite. > Cutting / Grinding on old Support.. > Site Inspektor to Perform MPI. > Tack welds new Pipe Support and QC to verify that Support are in the right position before welding. > Weld new Pipe Support. > Tidy up workplace and remove materials. Fireguard must have done fire guard training and had familiarization with the worksite and gas monitoring equipment. Fireguard to be present and in radio contact with CCR, during Hot Work. Radio contact to CCR channel 3

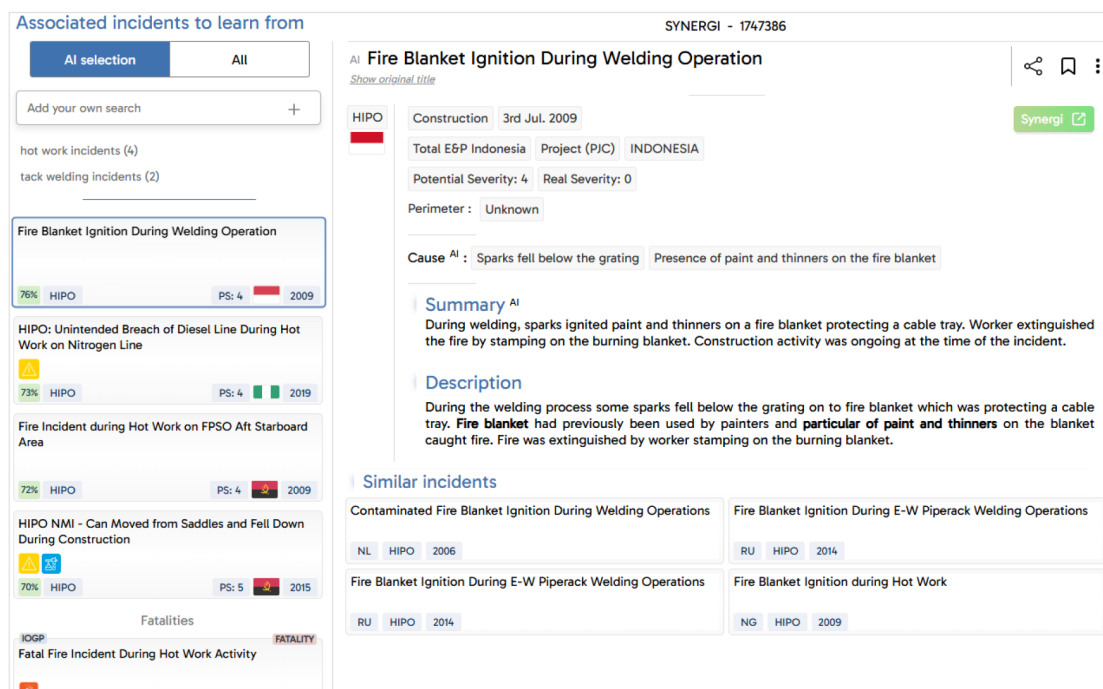


Figure 10—Result of a permit processed: After having selected their permit the user gets a selection of relevant past incidents.

The permit processing workflow – shown in Figure 11 – uses a chain of two LLMs along with the SafeTE search tool from Use Case 1:

- One LLM-based inference of context from work permit descriptions, used to generate precise and relevant search queries.
- One LLM-based agent for post-selection of search results, which revalidates the initial semantic search layer to ensure alignment with the intended context. This step is crucial to avoid cross-trade mismatches. For example, in the case of a well intervention job, incidents that occurred in a process facility—though semantically similar—would typically be considered irrelevant.

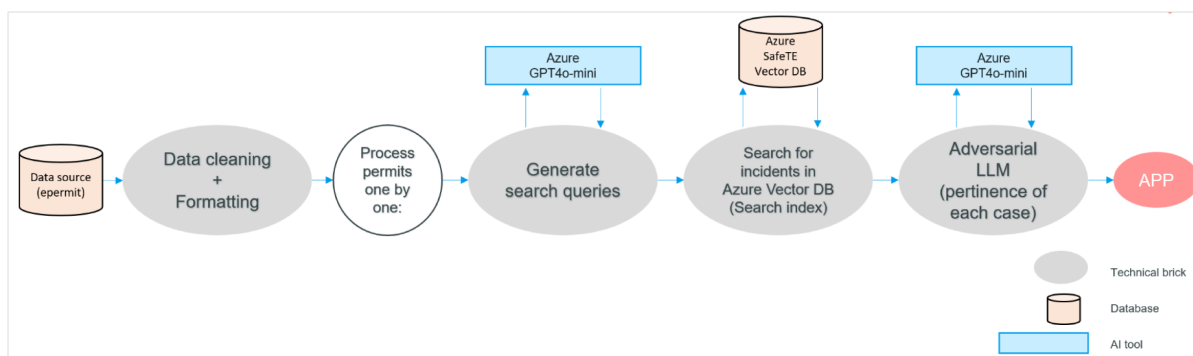


Figure 11—SafeWork: Processing of the new permits with 2 LLM to generate queries and verify the pertinence of our results

Use Case 3: Topic modelling on observation cards and safety-oriented insights

This module called *MySafety – Observation Cards* focuses on extracting structured insights from large volumes of observation cards, helping safety teams identify trends, recurring issues, and areas for improvement.

An initial overview presents the 37 categories of observation cards – identified using a clustering algorithm – organized by location (see some of them on Figure 12). This clustering proposal is obviously adapted to the typology of the assets supported, an offshore asset will see appearing cluster like "Road Traffic" or "excavation" while North Sea offshore asset could see a more stronger "Corrosion" category or "adverse weather".

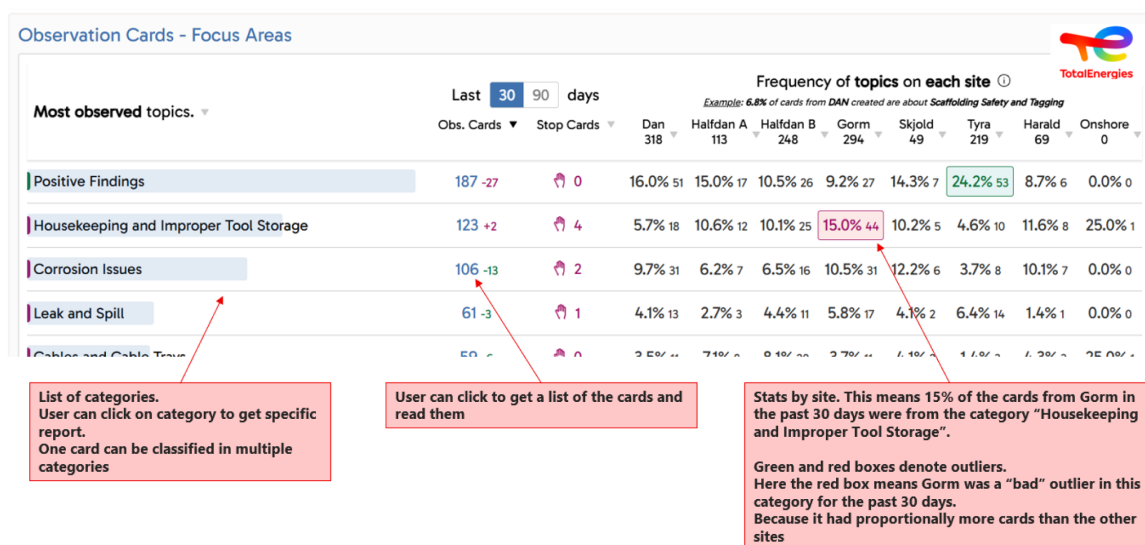


Figure 12—Layout of the MySafety Observation Cards tool

Each category can be clicked to get a detailed report (see next figure):

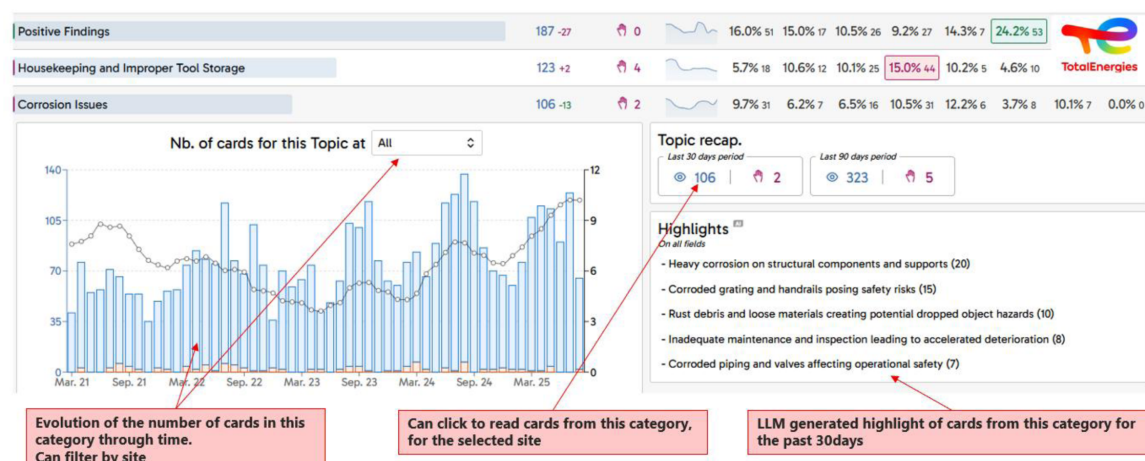


Figure 13—Detailed report for a category in the Observation Card tool.

This module employs various techniques, ranging from the creation of the 37 classes to the classification process itself (see Figure 14):

- Algorithmic topic modeling and clustering, enabling multi-class classification of observations based on thematic content
- LLM-based quantitative summarization of topic clusters, providing digestible overviews of key themes and trends
- Duplicate detection using semantic similarity, combined with time and location metadata, to identify when multiple individuals report the same situation

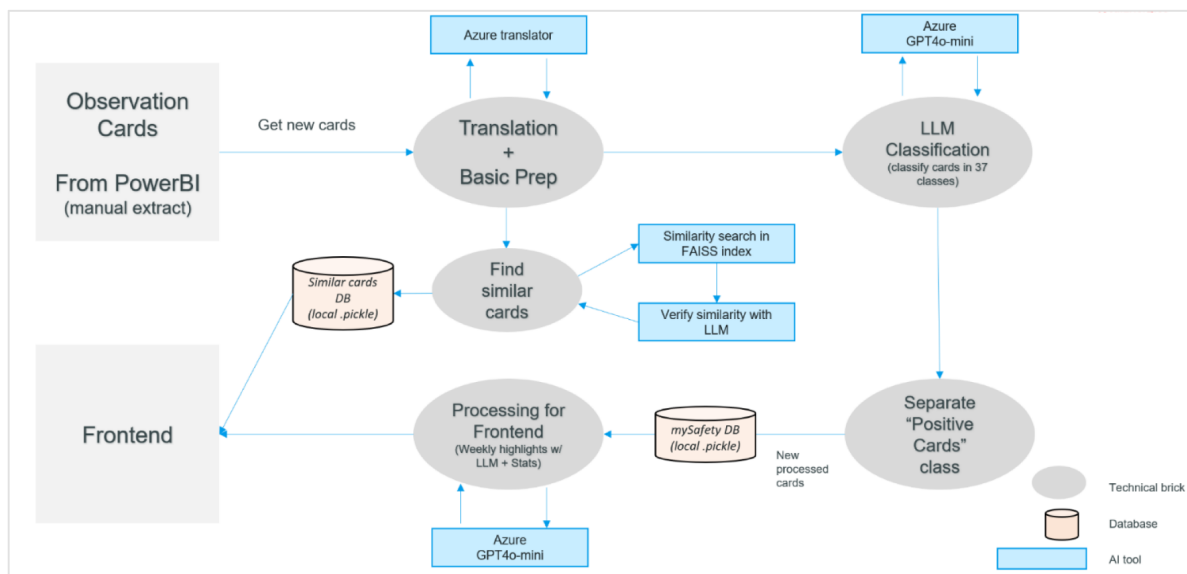


Figure 14—MySafety Observation Cards: processing and classification of new observation cards with LLM.

STEP3: Quality control

Ensuring quality is a critical component of any AI-driven solution. The SafeTE tool incorporates both upfront quality control and continuous user feedback mechanisms:

- Upfront quality control and testing: Benchmarks play a key role in maintaining high standards and tracking the impact of regularly updated versions of available LLMs. These benchmarks help ensure consistent performance and build trust in the system's outputs.
- User feedback collection: Throughout the tool, users have access to the original source material used by the AI to generate its conclusions. This transparency allows users to validate outputs and provide feedback, which can be integrated into the system to improve relevance and accuracy over time.

As an example, you can find in [Figure 15](#) the benchmark made for the classification of observation cards by LLM in use case 2. For each card, the model provides a list of classes. Such benchmarks are done by subject matter expert users, who evaluate the performance of each model. The different versions correspond either to new LLM version (or supplier) or to new peri-LLM algorithmic approaches. The set of questions and case studied is maintained over time.

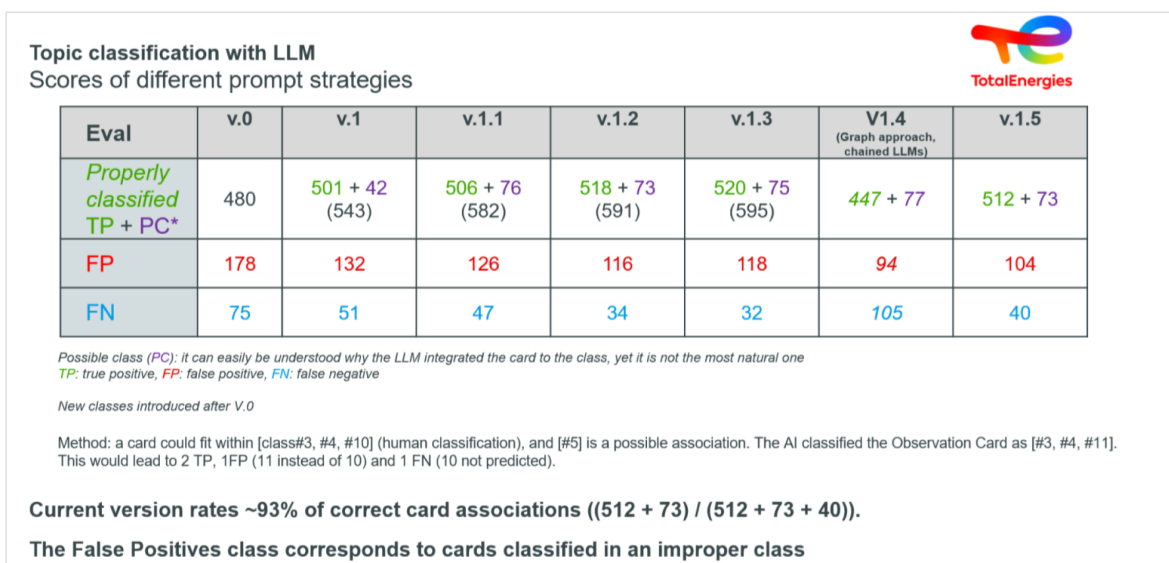


Figure 15—Benchmark for the observation cards classification. Done over a 400+ human evaluated testing set. Each card can have multiple class. Possible Class (in purple) were introduced

Data privacy and security aspects

Data privacy and security lead to frequent questions:

- How does the solution deals with LLM/AI and private proprietary data? Answer: The solution is based on an internal infrastructure (the digital platform) provided by the company that handles those aspects and does not expose our data to AI tenants that are not conform to the company policy. It can be mentioned for companies that would not accept any cloud tenant that we tested and ran successfully versions of the tool using open-source LLM deployed on premise (on high performance computers). Although those models are not always as performant as the proprietary ones, the difference identified on benchmarks tended to reduce over time for the given use cases, and they could be acceptable alternative to cloud solutions.
- How do you deal with potentially personal data? Answer: this solution is at high level a system promoting the exposure of data from other databases. As personal data and privacy is properly managed in the input database systems (no mention of names or personal data exist in those databases, or details are scrubbed details whenever required), this aspect did not prove to be a concern when the compliance evaluations were completed at design and execution phase.

Conclusion

The project illustrated in this SPE paper is a demonstration that Generative AI can contribute to enhance significantly the user experience when delivering appropriate and relevant insights on occupational safety.

When paired with thoughtful user interface design, the solution effectively handles the dilemma information overload – increase exposure, mitigates the subjectivity of incident recall by reducing reliance on individual memory and experience (Use Cases 1 and 2), and enables continuous attention on the analysis of daily-updated operational datasets (Use Case 3).

While the implementation using modern LLM APIs is relatively straightforward, the true complexity lies in crafting the right user experience—supported by iterative user feedback and ensuring robust ModelOps. This includes monitoring the performance of multiple models through benchmarks and statistical feedback collection.

Next steps involve adapting the tool to new use cases to process safety cases and scaling the tool to other locations (in progress, intrinsically easy as the solutions it brings are not geographically specific.)

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